ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

S.V.ISWARYA19
2513032

C02-3

EXPERIMENT 8

Aim:

To write program for determination and plotting of Magnetic Field Intensity, Intrinsic Impedance and Phase Velocity in different medium using MATLAB

Software required:

• MATLAB (R2018 or later)

Test case (i) Determination and plotting of Magnetic Field Intensity due to infinite long conductor at a distance r

```
clc;

clear;

I = input('Enter the current (A):');

r = input('Enter the distance (m):');

mu0 = 4*pi*1e-7;

H = I/(2*pi*r);

disp('Magnetic Field Intensity is');

disp(H);

% Plot x = linspace(0.1,10,100);

H_plot = I./(2*pi*x);

figure;

plot(x,H_plot,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Magnetic Field Intensity (A/m)');

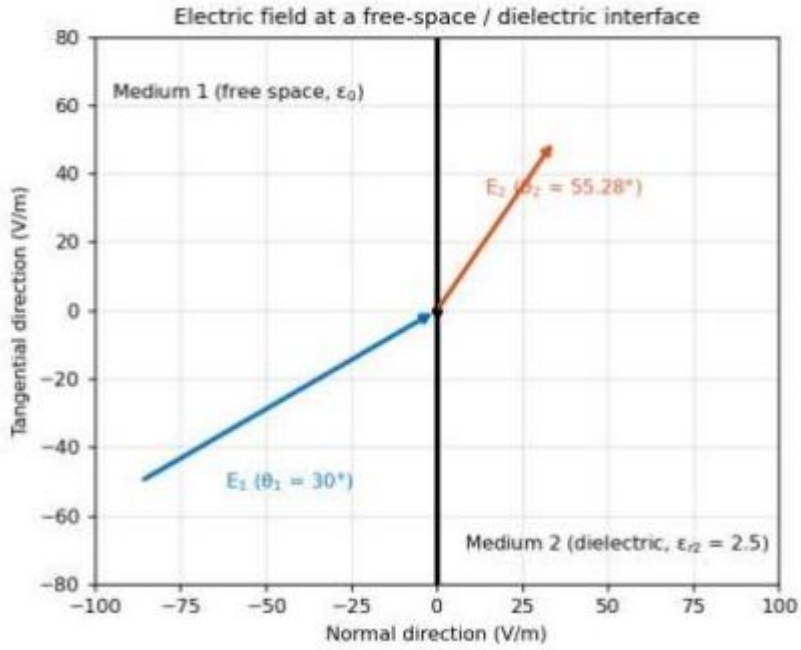
title('Magnetic Field Intensity vs Distance');
```

Output:

Enter the current (A): 5

Enter the distance (m): 1

Magnetic Field Intensity is 0.7958 A/m



Test case

(ii) Determination and plotting of Magnetic Flux Density in free space

```
clc;

I = 5;

mu0 = 4pi1e-7;

r = linspace(0.1,10,100);

H = I./(2pir);

B = mu0 * H;

plot(r,B,'LineWidth',2);

xlabel('Distance (m)');
```

```
ylabel('Magnetic Flux Density (T)');

title('Magnetic Flux Density vs Distance');
```

OUTPUT:

• Flux density decreases with distance

• Same inverse relation as magnetic field

```
Enter normal E-field in free space E1n (V/m): 86.60
Enter measured normal E-field in the dielectric E2n (V/m): 34.20
Enter reference relative permittivity: 2.5
Calculated relative permittivity = 2.5322
Percentage deviation = 1.29 %
>>
```

Test case (iii)

Determination and plotting of intrinsic impedance for different media

```
clc;

mu0 = 4pi1e-7;

epsilon0 = 8.854e-12;

er = linspace(1,5,100);

ur = linspace(1,2,100);

eta = sqrt((mu0.*ur)./(epsilon0.*er));

plot(er,eta,'LineWidth',2);

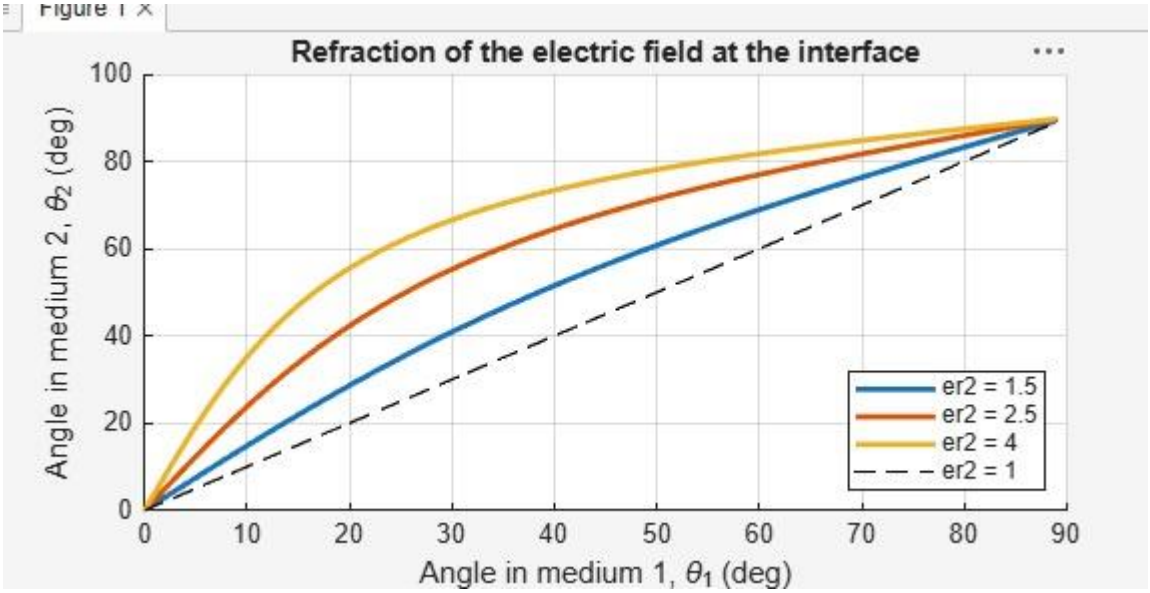
xlabel('Relative permittivity');

ylabel('Intrinsic Impedance (Ohms)');

title('Intrinsic Impedance vs Permittivity');
```

Output:-

• Impedance decreases with increasing permittivity



Test case (iv)

Determination and plotting of capacitance variation with dielectric constant (Glass)

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4; d = 1e-3;
er = linspace(4,8,100);
C = (epsilon0 .* er .* A) ./ d;
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Glass)');
```

Output:

Higher dielectric → higher capacitance

Test case (iv) Determination and plotting of phase velocity in different medium

```
clc;
mu0 = 4pi1e-7;
epsilon0 = 8.854e-12;
```

```
er = linspace(1,5,100);

ur = 1;

v = 1./sqrt(mu0*epsilon0*er*ur);

plot(er,v,'LineWidth',2);

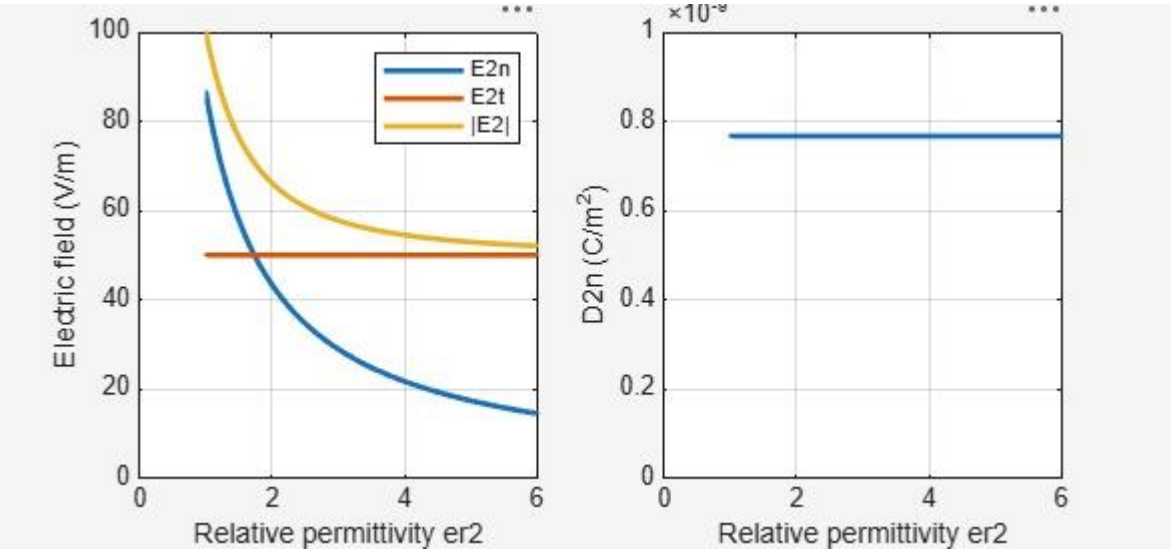
xlabel('Relative permittivity');

ylabel('Phase velocity (m/s)');

title('Phase Velocity vs Permittivity');
```

OUTPUT:

Velocity decreases with increase in permittivity.



RESULTS:

Thus the program was verified for magnetic field intensity, intrinsic impedance and phase velocity in different medium using MATLAB was successfully

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Set capacitor geometry and material properties	Set capacitor geometry and material properties	Correctly enters plate area, separation and relative permittivity with SI units.	Inputs are correct with a minor unit/ property issue.	Several inputs need correction.	Geometry or dielectric inputs are invalid.	25	2.5	
2	Calculate capacitance in MATLAB	Calculate capacitance in MATLAB	Uses the correct capacitance relation and calculates values for all specified media.	Calculations are mostly correct with minor errors.	Only some cases are calculated correctly.	Capacitance calculations are incorrect.	25	2.5	
3	Compare dielectric and geometry effects	Compare dielectric and geometry effects	Correctly evaluates trends versus permittivity/area/ separation and validates the model.	Trend and validation are mostly correct with minor omissions.	Trend discussion or validation is incomplete.	Incorrectly interprets parameter effects.	30	3	
4	Document engineering observations	Document engineering observations	Provides a complete table/plot with units and explains dielectric influence.	Documentation and explanation are mostly complete.	Documentation and explanation are mostly complete.	Results are incomplete or unexplained.	20	2	
Total							100	10	